7TH GRADE 3RD TRIMESTER EXAM STUDY GUIDE

Page 1 – Living Things, Classification, and Ecology

1 Characteristics of Living Things

- Living things are made of __cells__________.
- Living things maintain homeostasis____ to keep internal conditions stable.
- Living things respond to stimuli_________ in their environment.
- DNA contains the instructions for traits______.

2 Classification

- The science of classification is called taxonomy______.
- The three domains are:

Archaea

Bacteria

Eukarya

Complete the Linnaean Classification System:

1. Kingdom	→	Example: Animalia
2. Phylum	→	Chordata
3. Class	→	Mammalia
4. Order	→	Primates
5. Family	→	Hominidae
6. Genus	→	Homo
7. Species	→	sapiens

- A scientific name has two parts: Genus and species.

3 Ecology

A. Relationships

- Types (ex: predation, competition, mutualism): predation, competition, mutualism, commensalism, parasitism
- Example of each type: Predation: wolf eats deer; competition: plants compete for sunlight; mutualism: bee + flower; commensalism: bird nest in tree; parasitism: tick on dog.

B. Roles in an Ecosystem

- Producers: make their own food using sunlight
- Consumers (primary, secondary, tertiary): eat other organisms for energy
- Decomposers: break down dead matter; recycle nutrients

C. Biomes – Describe each biome and give two adaptations that would help an organism survive there.

Biome	General Description	Adaptations (2)
Rainforest	warm, wet; high biodiversity	1. broad leaves 2. camouflage
Desert	dry; little rainfall	1. store water 2. nocturnal
Tundra	very cold; frozen soil	1. thick fur 2. low growth
Temperate Deciduous Forest	four seasons; trees lose leaves	1. hibernate/migrate 2. leaf drop

D. Ecological Levels – List from smallest to largest.

Organism → Population → Community → Ecosystem → Biome → Biosphere →

E. Human Impacts

- Positive impacts: conservation, recycling, habitat restoration
- Negative impacts: pollution, deforestation, overuse of resources
- Give two examples of human activities that harm ecosystems: 1. deforestation 2. burning fossil fuels

F. Adaptations

- Structural (physical): body part/feature (thick fur, beak)
- Behavioral: action/behavior (migration, hibernation)
- Physiological: internal process (conserves water)

Page 2 – Photosynthesis, Respiration, and Matter Cycles

1 Photosynthesis

- Photosynthesis takes place in the **chloroplasts**.
- The green pigment that absorbs sunlight is **chlorophyll**.
- Reactants of photosynthesis:

CO_2 + **water** + **sunlight**

- Products of photosynthesis:

glucose + **oxygen**

Photosynthesis – Factors That Increase Rate

- Light intensity: higher = faster
- CO_2 concentration: higher = faster
- Temperature: optimal warm temp
- Water availability: more water = faster

Why Photosynthesis Slows in the Fall

- Days get shorter, temperatures drop, and **sunlight** decreases. Trees break down **chlorophyll**, so leaves change color and may fall.

2 Cellular Respiration

- Cellular respiration takes place in the **mitochondria**.
- The purpose of cellular respiration is to release **ATP/energy**.
- Types of cellular respiration:
  - Aerobic respiration (uses O_2): with oxygen; more ATP
  - Anaerobic respiration (no O_2): without oxygen; less ATP
- Reactants of respiration:

glucose + **oxygen**

- Products of respiration:

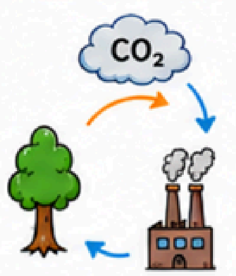
carbon dioxide + **water** + **energy (ATP)**

3 Matter Cycles

Carbon Cycle

- Photosynthesis removes **CO_2** from the atmosphere.
- Combustion releases **CO_2** into the atmosphere.
- Three carbon reservoirs:

atmosphere | oceans | fossil fuels



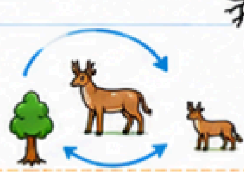
Nitrogen Cycle

- Nitrogen fixation is mainly performed by **bacteria**.
- Nitrogen is important because organisms use it to build proteins, DNA, and cells.

proteins | DNA | cells

Oxygen Cycle

- Photosynthesis releases O_2.
- Respiration and decomposition use O_2 and release
- O_2 is cycled between living things and the environment.



4 Conservation of Matter and Energy

- Matter and energy cannot be **created** or **destroyed**.
- Energy **flows** through ecosystems while matter **cycles** through ecosystems.

5 Draw and Label

Example answer:

Draw a simple food chain below with an apex predator at the end.

Producer → Primary consumer → Secondary consumer → Apex predator

Grass → **Rabbit** → **Snake** → **Hawk**